

The expected characteristic polynomial of a random lift of a  
cycle:  
an elementary proof, a generating function,  
and a machine-checked formalization

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**Abstract**

Let  $\Phi_{n,r}(x)$  denote the expected characteristic polynomial of a uniformly random permutation  $r$ -lift of the cycle  $C_n$ . We give a short, self-contained derivation of the closed form  $\Phi_{n,r}(x) = \chi_{C_n}(x) \cdot U_{r-1}(T_n(x/2))$ , where  $T, U$  are Chebyshev polynomials and  $\chi_{C_n}(x) = 2T_n(x/2) - 2$ ; equivalently, the  $d$ -matching polynomial of  $C_n$  (Hall–Puder–Sawin) is  $U_d(T_n(x/2))$ . This statement is equivalent, via the Hall–Puder–Sawin identity, to a conjecture of Hall proved combinatorially by Cochran, Groothuis, Herring, Rohatgi and Stucky (2018); our proof is independent, passing through the exponential formula and a Chebyshev generating function, and yields as byproducts the generating function  $\sum_{d \geq 0} \mu_{d,C_n}(x) z^d = (1 - 2T_n(x/2)z + z^2)^{-1}$ , a Fibonacci–Lucas evaluation, and a parity theorem for the quotient. The entire derivation from cycle weights to the closed form — including the exponential formula for  $S_r$  — is formalized in Lean 4 over Mathlib, with no axioms beyond Lean’s standard three.

## 1 Introduction and statement

The expected characteristic polynomial of a random lift is the object at the heart of the interlacing-families method of Marcus–Spielman–Srivastava [4], who used the  $r = 2$  (signing) case to prove that bipartite Ramanujan graphs exist in every degree; Hall–Puder–Sawin [2] extended the framework to arbitrary coverings. We evaluate it in closed form for the cycle.

A *permutation  $r$ -lift* of a graph  $G$  replaces each vertex by  $r$  copies and each edge  $\{u, v\}$  by the perfect matching  $u_i \sim v_{\sigma(i)}$  of a permutation  $\sigma \in S_r$  chosen independently and uniformly per edge. For the cycle  $C_n$ , gauge-fixing a spanning path reduces the lift to a single permutation  $\sigma \in S_r$  on the closing edge, and the lift decomposes into disjoint cycles: each  $\ell$ -cycle of  $\sigma$  contributes a cycle  $C_{n\ell}$  (fixed points contribute copies of  $C_n$ ). Since  $\chi_{C_m}(x) = \det(xI - A_{C_m}) = 2T_m(x/2) - 2$ , the expected characteristic polynomial is the expected cycle weight

$$\Phi_{n,r}(x) = \mathbb{E}_{\sigma \in S_r} \prod_{\ell \in \text{cycles}(\sigma)} (2T_{n\ell}(x/2) - 2). \quad (1)$$

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**Theorem 1.** For all  $n \geq 1$ ,  $r \geq 1$ , with  $Y = T_n(x/2)$ ,

$$\Phi_{n,r}(x) = U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) = \underbrace{(2T_n(x/2) - 2)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}(T_n(x/2))}_{\Psi_r(x)}.$$

The factored form identifies the quotient  $\Psi_r$  — by the theorem of Hall–Puder–Sawin [2] equating the expected new-eigenvalue characteristic polynomial with the  $d$ -matching polynomial ( $d = r - 1$ ) — as

$$\mu_{d,C_n}(x) = U_d(T_n(x/2)),$$

which is equivalent (through  $\mathcal{P}_d(2t) = U_d(t)$ ,  $\mathcal{C}_n(2t) = 2T_n(t)$  and the composition identity  $U_{dn+n-1} = U_d(T_n)U_{n-1}$ ) to *Hall's conjecture*  $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$ , proved in [3]. Our route is different and elementary; we make no use of matching-polynomial combinatorics.

**Corollary 2.**  $\sum_{d \geq 0} \mu_{d,C_n}(x) z^d = \frac{1}{1 - 2T_n(x/2)z + z^2}, \quad \text{and} \quad \sum_{r \geq 0} \Phi_{n,r}(x) z^r = \frac{(1 - z)^2}{1 - 2T_n(x/2)z + z^2}.$

**Corollary 3** (real-rootedness). All roots of  $\Phi_{n,r}$  lie in  $[-2, 2]$ : with  $x = 2 \cos \theta$ ,  $\chi_{C_n}$  vanishes at  $\theta = 2\pi m/n$  and  $\Psi_r = \sin(rn\theta)/\sin(n\theta)$  vanishes at  $\theta = m\pi/(rn)$  with  $r \nmid m$  (when  $r \mid m$  the denominator vanishes too and  $\Psi_r = \pm r \neq 0$ ); this leaves exactly  $n(r - 1)$  roots, matching  $\deg \Psi_r$ .

**Corollary 4** (Fibonacci–Lucas evaluation).  $\Phi_{n,r}(3) = (L_{2n} - 2) F_{2nr}/F_{2n}$ , since  $T_n(3/2) = L_{2n}/2$  and  $U_{r-1}(L_{2n}/2) = F_{2nr}/F_{2n}$ .

**Corollary 5** (parity).  $\Psi_r(-x) = (-1)^{n(r-1)}\Psi_r(x)$ : the quotient is supported on exponents of a single parity, like a matching polynomial. (For bipartite  $C_n$  it is an even polynomial for every  $r$ ; at  $r = 2$  it is the classical matching polynomial  $2T_n(x/2)$  of  $C_n$ , consistent with *Godsil–Gutman* [1].)

## 2 Proof

Write  $f(\ell) = 2T_\ell(Y) - 2$  with  $Y = T_n(x/2)$ ; by the composition identity  $T_{n\ell} = T_\ell \circ T_n$  this is the per-cycle factor in (1).

*Step 1 (exponential formula).* For any commutative ring  $R$  and  $f : \mathbb{N} \rightarrow R$ , the  $S_r$ -average  $\Phi_r = \mathbb{E}_\sigma \prod_{\ell \in \text{cycles}(\sigma)} f(\ell)$  satisfies

$$r \Phi_r = \sum_{k=1}^r f(k) \Phi_{r-k}, \quad \Phi_0 = 1, \tag{2}$$

the coefficient form of  $\sum_r \Phi_r z^r = \exp(\sum_\ell f(\ell) z^\ell / \ell)$ . (Proof: group  $\sigma$  by cycle type  $\lambda$  with Cauchy's count  $r!/z_\lambda$ , mark a part, and re-index; see the Lean file `Paper2ExpFormula.lean` for the fully detailed version.)

*Step 2 (Chebyshev generating function).* With  $f(\ell) = 2T_\ell(Y) - 2$ ,

$$\sum_{\ell \geq 1} f(\ell) \frac{z^\ell}{\ell} = -\log(1 - 2Yz + z^2) + 2\log(1 - z) \implies \sum_{r \geq 0} \Phi_r z^r = \frac{(1 - z)^2}{1 - 2Yz + z^2}.$$

Extracting coefficients against  $\sum U_r(Y)z^r = (1 - 2Yz + z^2)^{-1}$  gives  $\Phi_r = U_r - 2U_{r-1} + U_{r-2}$ , and the three-term recurrence  $U_r + U_{r-2} = 2YU_{r-1}$  collapses this to  $(2Y - 2)U_{r-1}(Y)$ .  $\square$

In the formalization the analytic Step 2 is replaced by an equivalent finite argument: the closed form is shown to satisfy (2) directly (a second-difference induction), and (2) with  $\Phi_0 = 1$  determines  $\Phi$  over a characteristic-zero domain.

### 3 The formalization

The core algebraic derivation — the exponential formula, the closed form of Theorem 1 for all  $n, r$  (equivalently, everything from the cycle weights (1) to the closed form), its factorization, and the parity Corollary 5 — is machine-checked in Lean 4 over Mathlib (v4.30). The main entry points:

- `Paper2.expFormula` — the exponential formula (2) in integer form  $r A_r = \sum_k r^{(k)} f(k) A_{r-k}$  over any commutative ring, proved via Mathlib’s Cauchy cycle-type count (`Equiv.Perm.card_of_cycleType`);
- `Paper2.thm1_Sr` — Theorem 1 in the form: the expected cycle weight (1) equals the closed form, for all  $n, r$ , over any  $\mathbb{Q}$ -algebra domain;
- `Paper2.cG_factored`, `Paper2.quotient_parity` — the factorization and Corollary 5.

The three remaining corollaries are formalized only in part: Corollary 2’s generating function appears as the finite recurrence (2) rather than a literal power-series identity; of Corollary 3 only the underlying trigonometric factorization (`cor1_factored`) is checked, not the “all roots in  $[-2, 2]$ ” statement end-to-end; and Corollary 4 (`cor2_phi3`) is a bounded `native_decide` over  $n, r \leq 6$ .

Trust base for the general theorems: `propext`, `Classical.choice`, `Quot.sound` only (checked with `#print axioms`); the bounded `native_decide` checks additionally invoke `Lean.ofReduceBool`. No `sorry`. The single remaining unformalized ingredient of the *graph-theoretic* statement is the sentence before (1): a lift of a cycle is a disjoint union of cycles — elementary covering-space combinatorics for which Mathlib currently lacks the vocabulary.

*Remark 6.* Everything here is stated for the uniform permutation model; by [2] the quotient coincides with the  $(r - 1)$ -matching polynomial, so Theorem 1 can alternatively be deduced from [3]. We believe the generating function of Corollary 2, the evaluation of Corollary 4, and the formalization are new.

### References

- [1] C. Godsil, I. Gutman, *On the matching polynomial of a graph*, in: Algebraic Methods in Graph Theory, 1981.
- [2] C. Hall, D. Puder, W. Sawin, *Ramanujan coverings of graphs*, Adv. Math. **323** (2018), 367–410.
- [3] G. Cochran, C. Groothuis, A. Herring, R. Rohatgi, E. Stucky, *A new (combinatorial) proof of the commutativity of matching polynomials for cycles*, arXiv:1810.05889 (2018).

- [4] A. Marcus, D. Spielman, N. Srivastava, *Interlacing families I: Bipartite Ramanujan graphs of all degrees*, Ann. of Math. **182** (2015), 307–325.